Quiz #8

1. For each situation, select the most appropriate method from the list below.

   - One-sample t-test
   - Independent two-sample t-test
   - Paired t-test
   - χ^2 goodness-of-fit test
   - χ^2 test for independence

   (a) Researchers take independent random samples of students from public and private universities. They want to determine whether the average number of hours worked per week differs between the two populations.

   (b) A survey records each person's marital status (never married, married, or previously married) and employment status (employed, unemployed, or outside the labor force). Researchers want to determine whether these two variables are associated.

   (c) Researchers select 400 families with two children. Both children take the same vocabulary test, and the researchers want to determine whether first-born children score higher than their younger siblings.

   (d) A county wants to determine whether the age distribution of people selected for jury duty is consistent with the known age distribution of eligible residents.

2. Determine whether each statement is True or False. Briefly explain your answer.

   (a) A fair coin is tossed 100 times. The standard error for the percentage of heads is 5 percentage points. Hint: The standard deviation for a single coin toss is $\sqrt{p(1-p)}$

   (b) In the same 100 tosses, the standard error for the difference between the percentage of heads and the percentage of tails is $\sqrt{5^2 + 5^2}$ percentage points.

   (c) To determine whether the age distribution of jurors differs from the county population, a separate z-test should be performed for each age category.

3. A polling organization selects a simple random sample of n voters from a population of N voters, without replacement. In polling, we often treat the sampled responses as independent and identically distributed (i.i.d.).

   (a) Are the sampled responses exactly independent? Briefly explain.

(b) For which poll is the independence assumption a more reasonable approximation? Explain your choice using n and N.

- Poll A: $n = 1{,}000$, $N = 1{,}000{,}000$
- Poll B: $n = 1{,}000$, $N = 5{,}000$

4. A materials laboratory compares two independent random samples of steel parts. Assume the sample sizes are large enough for the two-sample z approximation.

Process	Sample size	Mean strength	SD
A	100	520 MPa	30 MPa
B	100	510 MPa	40 MPa

Test $H_0 : \mu_A = \mu_B$ against $H_A : \mu_A \neq \mu_B$.

(a) Compute the estimated standard error of $\bar{X}_A - \bar{X}_B$.

(b) Compute the two-sample z statistic.

5. A manufacturing company tests whether a new assembly fixture reduces the mean time needed to assemble a device. Technicians are randomly assigned to two independent groups, and each uses only one fixture. Assume assembly times are approximately normal within both groups.

The assembly times (in minutes) are:

```
import numpy as np
from scipy import stats

new = np.array([18, 20, 19, 22, 17, 21, 20, 19])
standard = np.array([19, 21, 20, 22, 23, 20, 21, 22])

result = stats.ttest_ind(new, standard,
                         equal_var=False,
                         alternative="less")
print(result.statistic, result.pvalue)
```

(a) State the null and alternative hypotheses tested by this code.

(b) Explain what `ttest_ind`, `equal_var=False`, and `alternative="less"` mean. Why would `stats.ttest_rel()` be inappropriate?

(c) The code returns approximately $t = -2.05$ and $P = 0.030$. What is your conclusion at $\alpha = 0.05$? Does statistical significance alone establish that the time saving is important in practice?